

ORACLE



MySQL Implementation Essentials Bootcamp

Workshop

Marco Carlessi

marco.carlessi@oracle.com

MySQL GBU

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Agenda



- Introduction
- Installation and Architecture
- MySQL Database Design
- MySQL Security
- MySQL Backup
- MySQL Replication
- MySQL High Availability
- MySQL Monitoring

Bootcamp Introduction

Workshop Overview



Goal

- Execute labs on based on MySQL 8.4: Essentials online course
 - <https://mylearn.oracle.com/ou/course/mysql-84-essentials/141332/>
 - Mandatory training to complete before attend

Bootcamp structure

- Introduction slides
- labs

Labs link

- <https://mc-carlessi.github.io/mysql-enterprise-security/workshops/student/index.html>

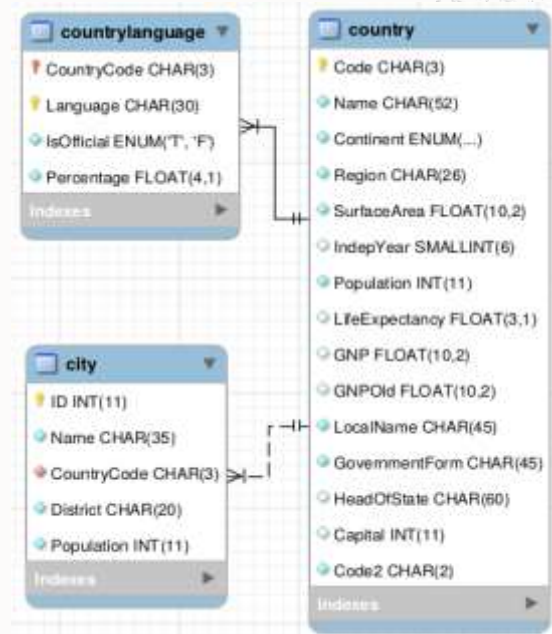
Timing

- 9:00 AM – 12:00 PM / 1:00 PM – 4:00 PM
 - 1 hour lunch break
 - A couple of coffee breaks



What is MySQL?

- Relational Database
 - Transactional, ACID
 - InnoDB storage engine: ARIES, MVCC
 - OLTP: low latency, high throughput
- SQL+NoSQL
 - NoSQL is JSON based document store
- Many options for DEVs and DBAs
 - CTE, Windows Functions, Replications for read scale out, Native High Availability, Native Disaster Recovery, and much more
- Simple, Solid, Secure
 - Easy to use, stable, fast, proven at scale



417 systems in ranking, February 2024

Rank			DBMS	Database Model	Score		
Feb 2024	Jan 2024	Feb 2023			Feb 2024	Jan 2024	Feb 2023
1.	1.	1.	Oracle 🏆	Relational, Multi-model 📄	1241.45	-6.05	-6.08
2.	2.	2.	MySQL 🏆	Relational, Multi-model 📄	1106.67	-16.79	-88.78
3.	3.	3.	Microsoft SQL Server 🏆	Relational, Multi-model 📄	853.57	-23.03	-75.52
4.	4.	4.	PostgreSQL 🏆	Relational, Multi-model 📄	629.41	-19.55	+12.90
5.	5.	5.	MongoDB 🏆	Document, Multi-model 📄	420.36	+2.88	-32.41

Source: <https://db-engines.com/en/ranking>



What's in it for partners?



- Leading database for Web (LAMP) and new standards (e.g. OpenStack)
- New markets, scale out quickly
- Great choice for projects where end customer TCO is critical
- High margins, high growth potential
- Ongoing subscription revenue, keep partners engaged long term
- Premier lifetime support, high customer satisfaction
- Will work alongside Oracle DB or SQL Server with projects
- You can provide professional services to configure and maintain MySQL Enterprise tools and features
- SQL Server and Oracle dba's can be easily trained

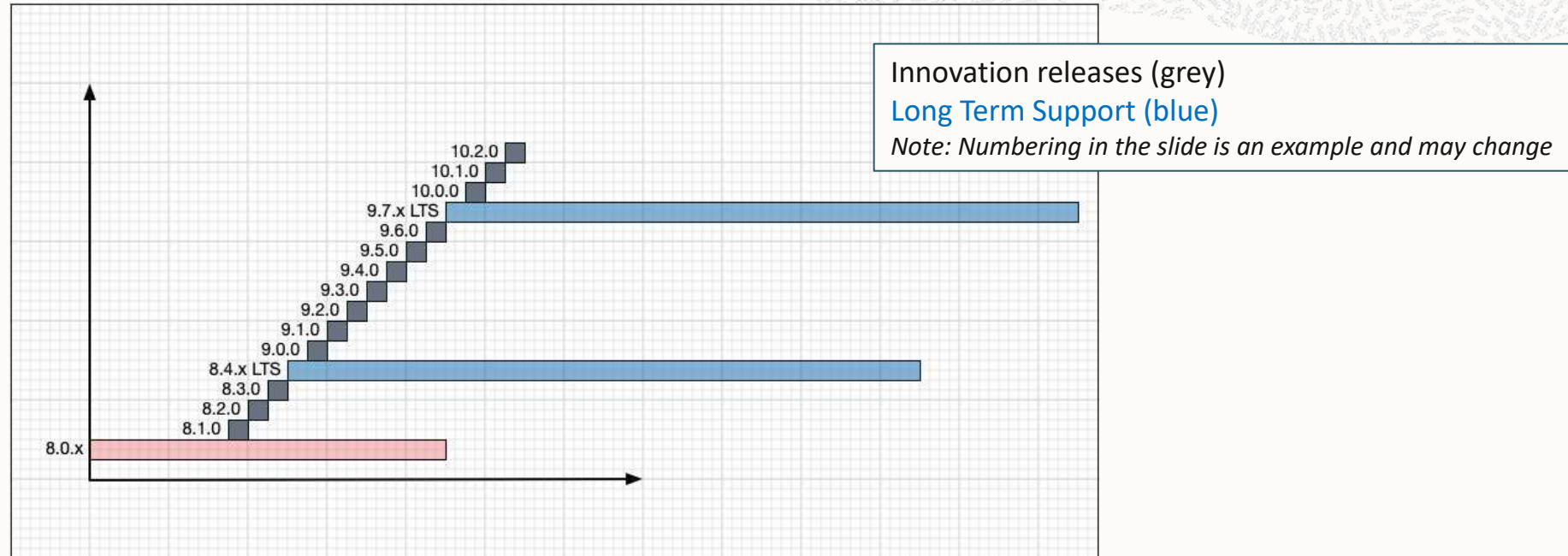
Bootcamp

- Focus on MySQL 8.4 Enterprise
- Each student works with 4 Linux servers
 - 1 application server and 3 MySQL Servers
 - Be careful with passwords, if you change what inside the lab description, it's your responsibility remember...
- Focus on operations, less DDL and DML



MySQL LTS & Innovation Releases

MySQL Community, Enterprise and HeatWave



MySQL Long-Term Support (LTS) Releases

- bugfix & security patches only
- backwards compatibility
- support lifecycle: 5y premier + 3y extended
- Include tools (like MySQL Shell)

MySQL Innovation Releases

- leading-edge innovations
- will likely released every quarter
- Include tools (like MySQL Shell)
- Connectors released as Innovation only

<https://blogs.oracle.com/mysql/post/introducing-mysql-innovation-and-longterm-support-lts-versions>



MySQL Installation and Architecture

MySQL – Installation

Windows

- **MySQL Installer**
 - Includes most MySQL Packages for Windows in one single package
 - Some MySQL apps rely on Microsoft .NET Framework 4.5.2 and Microsoft Visual C++ 2019 Redistributable Package
 - MySQL Database, MySQL Connectors, MySQL Workbench, mysqlslap (Load generation tool) and sample models
 - Documentation and Sample databases
 - First setup by wizard
- **Binary zip ("Noinstall Archive")**
 - just the binaries
 - manual configuration

Linux

- **Repositories for Community, packages (yum, rpm, deb) for Community and Enterprise**
 - Be careful: Some Linux distribution provide forks (like MariaDB), but it's easy to reinstall MySQL over MariaDB
 - Dependencies resolution
- **Binary tarball or zip**
 - For Community and Enterprise
 - Pay attention to dependencies, like
 - MySQL linux dependencies
 - libaio
 - /lib64/libtinfo.so.5 (usually contained in ncurses-compat-libs)

There is no best option between packages or zip/tar, it's a matter of knowledge, company standards, flexibility...

Package Manager for MySQL on Linux

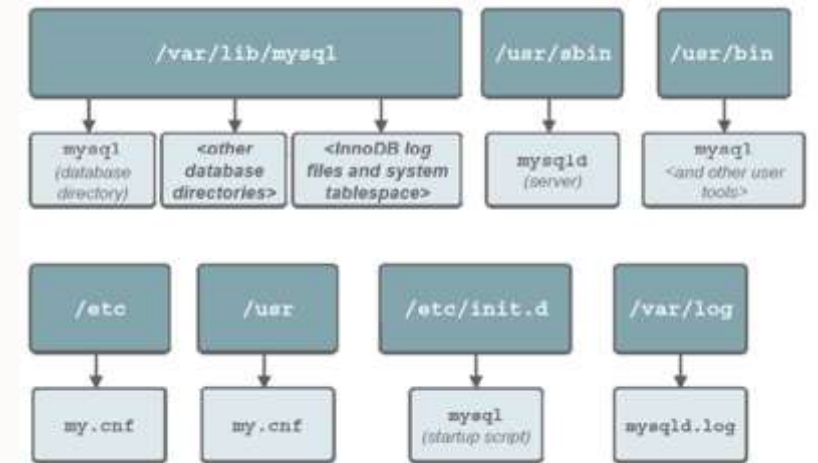
- Package managers simplify the MySQL installations, that also installs the dependencies
 - On RPM-based systems (Oracle Linux, Red Hat...), like

```
yum install mysql-community-server
```

```
yum install mysql-workbench
```
 - On APT-based systems (Ubuntu, Debian...)

```
apt-get install mysql-community-server
```

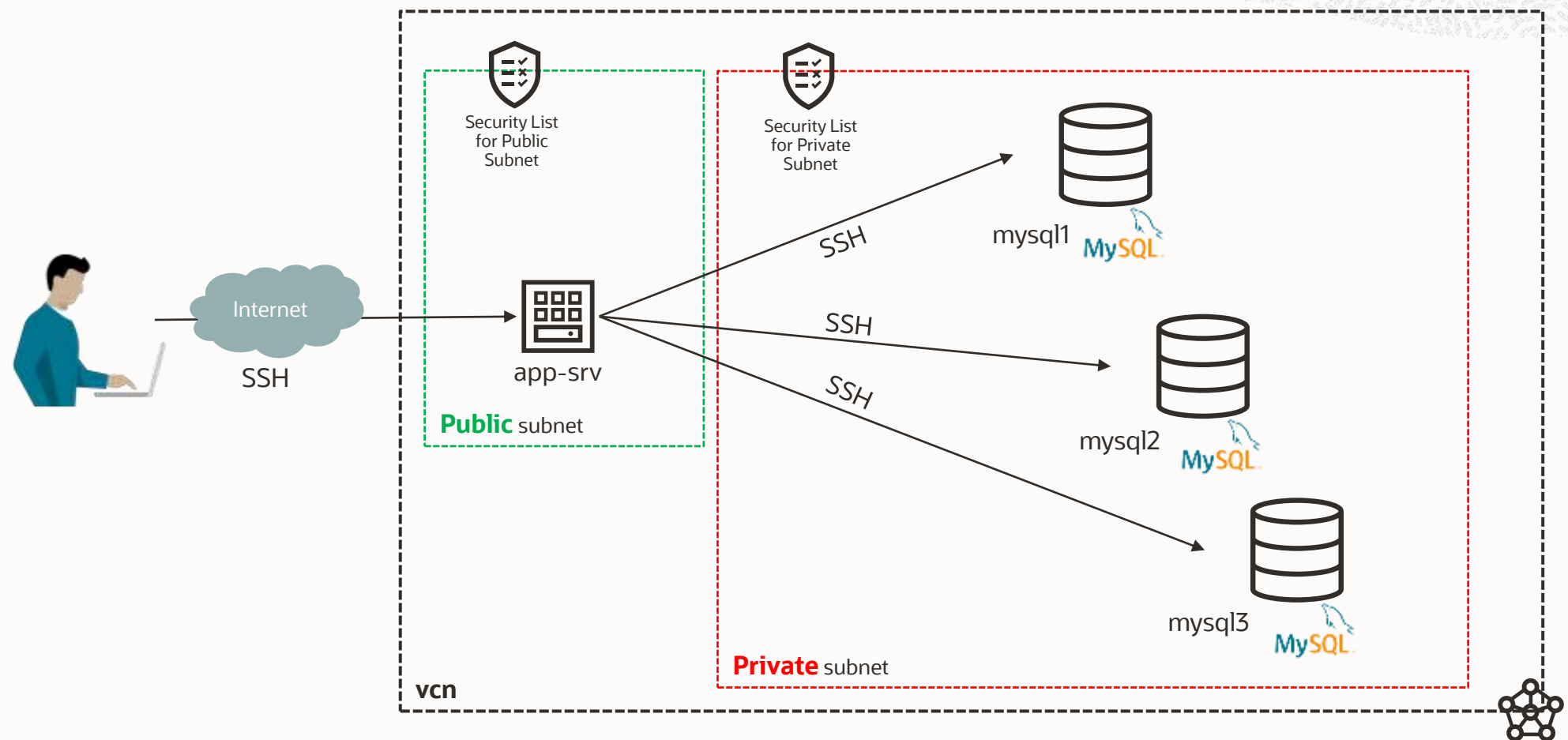
```
apt-get install mysql-workbench
```
- Package managers have their own defaults
 - Where are the configuration files
(/etc/my.cnf, /etc/mysqld/my.cnf, ...), log files, ...
- Enterprise Edition have no public repository, so download .rpm or .deb from <https://support.oracle.com> website
 - There is a dedicated package (MySQL Database/Components Yum Repository TAR) that helps setup of a self-hosted Yum repository



Default example for Oracle Linux

LAB 1

Our Lab



Each attendee has a dedicated environment, separated by the others



LAB 2a

MySQL Installation deep dive steps



- Basic steps are the same for all the platforms
 - we see examples for Linux
- MySQL installation steps
 - (optional) tune operating system settings
 - Create user that runs MySQL server process
 - Create directories infrastructure
 - Install the generic binaries distribution
 - Assign ownership and permission to directories and files
 - First setup of configuration file (my.cnf)
 - Install & Initialize the instance
 - Start mysql server
 - Configure automatic startup
 - Install tools, load plugins and configure what is needed for operation, security and Business Continuity
 - Backups, monitoring, security features, additional plugins...

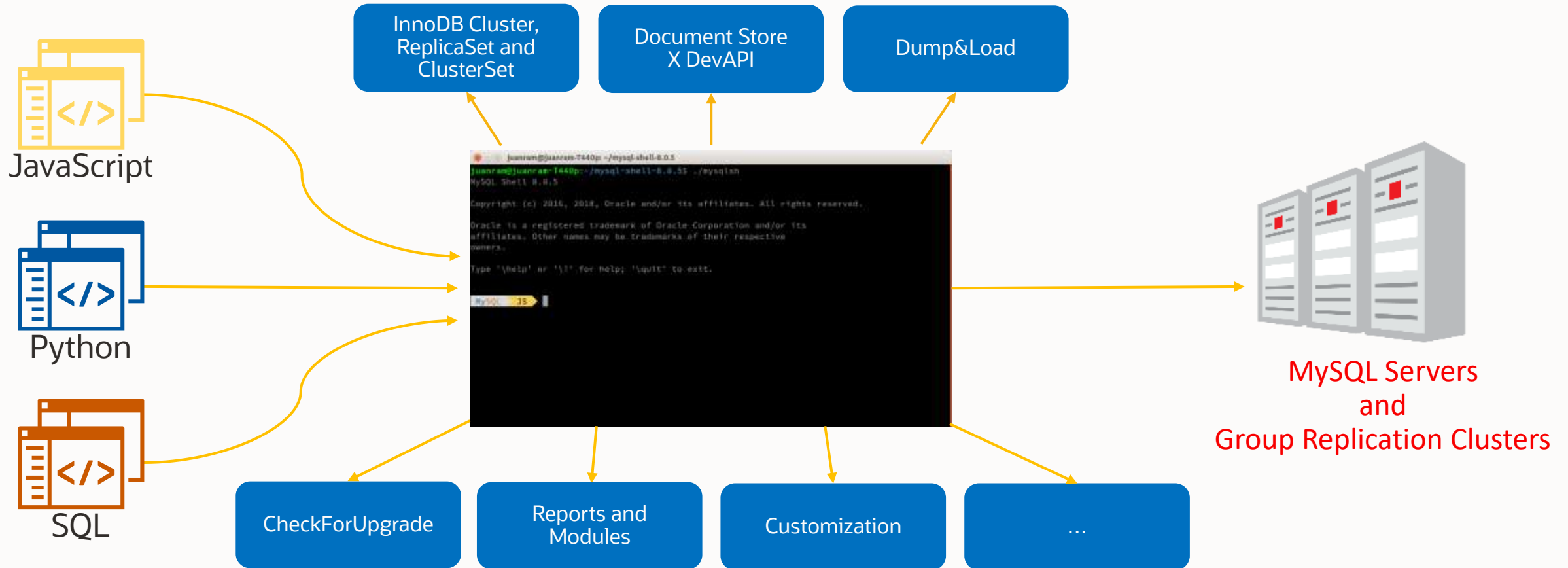
<https://dev.mysql.com/doc/refman/8.4/en/installing.html>

LAB 2b

LAB 2c

MySQL Shell

A single unified client for all administrative and operations tasks



<https://dev.mysql.com/doc/mysql-shell/8.4/en/>

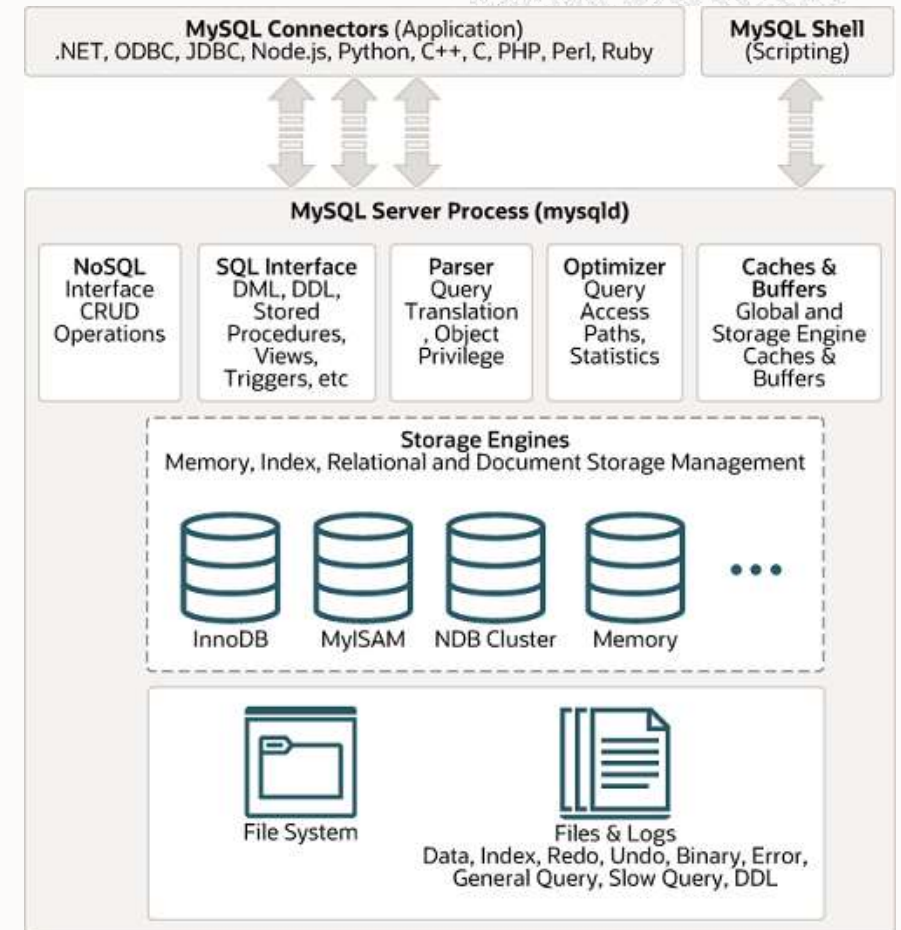


LAB 2d

MySQL Design

Storage Engines

- MySQL server architecture isolates the application programmer and DBA from all of the low-level implementation details at the storage level
- Some of the key feature differentiations include
 - Concurrency
 - Transaction Support
 - Referential Integrity
 - Physical Storage
 - Index Support
 - Memory Caches
 - Performance Aids
 - Miscellaneous Target Features
- Examples
 - InnoDB: recommended and default
 - MyISAM: small on disk, non transactional, no crash consistency, no online backup, ...



MySQL InnoDB



- InnoDB is a general-purpose storage engine that balances high reliability and high performance
- InnoDB is **ACID** compliant and **fully transactional** with ROLLBACK/COMMIT/Foreign Keys
- A typical MySQL database is normally in the lower three-digit GB
 - Theoretical limit of the maximum tablespace size is 64TB
 - It can serve more than 1000 columns per table
 - Contain a maximum of 64 secondary indexes
 - A row can have a size of up to 64KB (less BLOBs)
- InnoDB supports encryption and compression
- Online backups
- Multiple options for performance and tuning
- Built in High Availability and Disaster Recovery

MySQL Data Types

- Each data type has different ranges, defaults and storage requirements
- MySQL provides also dedicated functions
- Numeric exact value
 - INTEGER, INT, SMALLINT, TINYINT, MEDIUMINT, BIGINT
 - DECIMAL, NUMERIC
- Numeric approximate value
 - FLOAT, DOUBLE
- Bit-Value Type
 - BIT
- Date and Time Types
 - DATE, DATETIME, TIMESTAMP, YEAR
- String Types
 - CHAR, VARCHAR
 - BINARY, VARBINARY
 - ENUM
 - SET
- BLOB
 - TINYBLOB, BLOB, MEDIUMBLOB, and LONGBLOB
- TEXT
 - TINYTEXT, TEXT, MEDIUMTEXT, and LONGTEXT
- Spatial
 - GEOMETRY, POINT, LINESTRING, POLYGON
 - MULTIPOINT, MULTILINESTRING, MULTIPOLYGON, GEOMETRYCOLLECTION
- JSON Data Type

Type	Storage Required	Signed Range
FLOAT	4 bytes	-3.402823466E+38 to -1.175494351E-38 to 1.175494351E-38 to 3.402823466E+38
DOUBLE	8 bytes	-1.7976931348623157E+308 to -2.2250738585072014E-308 and 2.2250738585072014E-308 to 1.7976931348623157E+308

Value	CHAR (4)	Storage Required	VARCHAR (4)	Storage Required
' '	' '	4 bytes	' '	1 byte
'ab'	'ab '	4 bytes	'ab'	3 bytes
'abcd'	'abcd'	4 bytes	'abcd'	5 bytes
'abcdefgh'	'abcd'	4 bytes	'abcd'	5 bytes

MySQL Indexes

- Optimize the common lookup and DML operations

- BTREE
- HASH
- RTREE (spatial only)

- Automatic storage of data in a sorted order

- Non-Unique - May refer to many records
- Unique or Primary - Must refer to only one record

- Special cases

- Full-Text (CHAR, VARCHAR, or TEXT)
- Multi-value (JSON only)

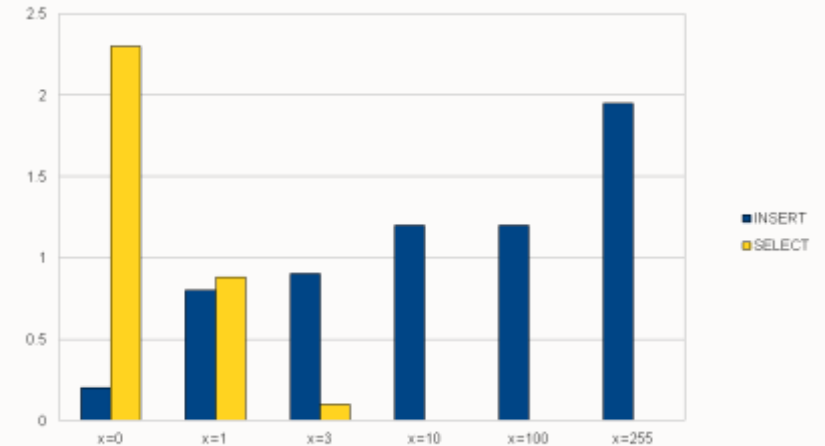
- Prefix indexes

- For string columns, indexes can be created that use only the leading part of column values
- `CREATE INDEX name ON customer(COL1(10));`

- Multiple-Column indexes

- can be considered a sorted array

```
index key123 (col1,col2,col3)
index key12 (col1,col2)    <- Not needed!
index key1 (col1)         <- Not needed!
```



LAB 3

Task 1

JSON data type



- JSON (JavaScript Object Notation) is an open standard file format that uses human-readable text
- Native JSON data type introduced with MySQL 5.7
 - Native internal binary format for efficient processing & storage
 - Up to 10x faster than storing as text
 - Provides Document Validation
 - `INSERT INTO employees VALUES ('some random text');`
 - `ERROR 3130 (22032): Invalid JSON text: "Expect a value here." at position 0 in value (or column) 'some random text'.`
- It use Objects and Arrays
- Its ease of use, speed of processing and easy integration made it extremely popular for Web-based applications
 - Simple data format that allows programmers to store and communicate sets of values
 - JSON's lack of a predefined schema makes it easy for developers to store and update documents
- Built-in JSON functions, for store, search, update, and manipulate Documents
- Indexing available via generated columns (automatically used by the optimizer)

```
"firstName":"John"

{"firstName":"John", "lastName":"Doe"}

"employess":[
  {"firstName":"John", "lastName":"Doe"}
  {"firstName":"Anna", "lastName":"Smith"}
  {"firstName":"Peter", "lastName":"Jones"}
]
```



JSON indexing with Generated Columns

- Generated Columns are computed from an expression included in the column definition

```
CREATE TABLE t1 (  
  id INT NOT NULL PRIMARY KEY auto_increment,  
  my_integer INT,  
  my_integer_plus_one INT AS (my_integer+1)  
);
```

id	my_integer	my_integer_plus_one
1	10	11
2	20	21

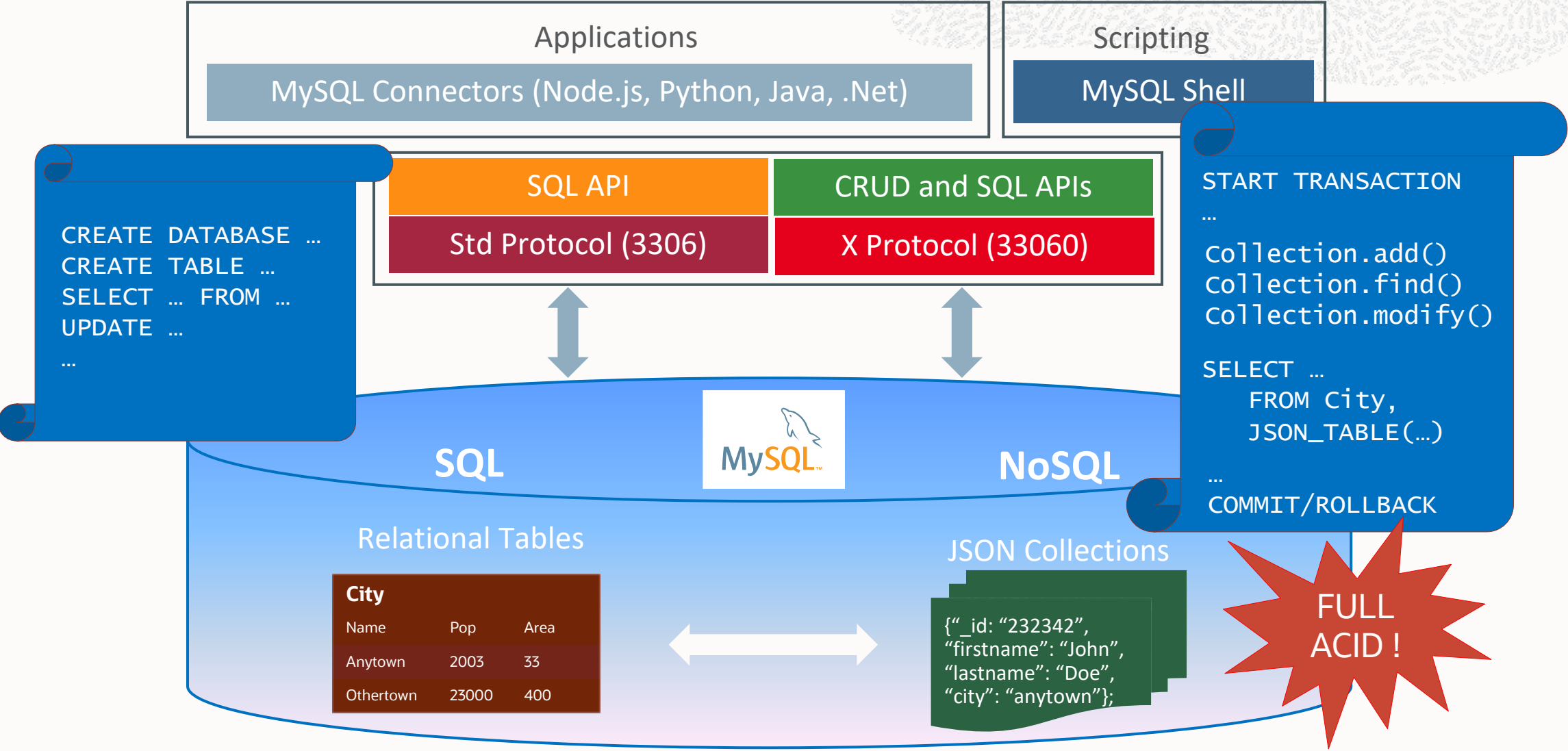
- Generated Columns may have indexes
- Example of JSON indexing

```
CREATE TABLE features (  
  id INT NOT NULL PRIMARY KEY auto_increment,  
  feature JSON  
);
```

```
ALTER TABLE features ADD feature_type VARCHAR(30) AS (feature->"$.type");  
ALTER TABLE features ADD INDEX (feature_type);
```

```
SELECT feature FROM features WHERE feature_type = "feature";  
SELECT feature FROM features WHERE feature->"$.type" = "feature";
```

MySQL Document Store: SQL+NoSQL database



LAB 3

Task 2 & Task3

MySQL Security

MySQL Database Hardening Best Practices



Installation

- Keep MySQL up to date
- MySQL Installer for Windows
- Yum/Apt/other Repositories

Authentication

- Password Policies
- Multi factor authentication
- External Authentication
- X.509

Authorization

- Remove Extra Accounts
- Grant Minimal Privileges
- Audit users and privileges

Configuration

- Firewall
- Auditing and Logging
- Limit Network Access
- Monitor changes

Encryption

- SSL/TLS for Secure Connections
- Data Encryption (AES, RSA)
- TDE
- Masking and De-Identification

Collateral

- Encrypt Backups
- Secure Replications
- Secure HA and DR

MySQL User Accounts



- Within MySQL, accounts are composed by three elements:
 - Username
 - blank usernames are considered anonymous
 - Host, the reference of the machine where the mysql client run
 - Used transparently by the client/connector
 - May use IP or FQDNs or domains with wildcards (e.g. '%' means every host)
 - Password
 - MySQL 8 default plugin is caching_sha2_password
 - MySQL 5.X format is mysql_native_password (it uses the SHA1 algorithm which NIST no longer recommend)
 - It's possible to have users with different password plugin in the same instance
 - Default plugin can be specified with the variable default_authentication_plugin

- Examples

```
CREATE USER 'username1'@'%' IDENTIFIED BY 'VeryComplex1!';  
CREATE USER 'username2'@'myhost.oracle.com' IDENTIFIED BY 'VeryComplex2!';  
CREATE USER 'username3'@'192.168.1.1' IDENTIFIED BY 'VeryComplex3!';  
CREATE USER ''@'%' IDENTIFIED BY 'VeryComplex4!';  
CREATE USER 'username4'@localhost IDENTIFIED WITH mysql_native_password BY 'VeryComplex4!';
```



Some Password Management features



- Password validation to enforce secure passwords
- Password expiration
 - To require passwords to be changed periodically
- Password reuse restrictions
 - To prevent old passwords from being chosen again
- Password verification
 - To require that password changes also specify the current password to be replaced
- Dual passwords
 - To enable clients to connect using either a primary or secondary password
- Password strength assessment
 - To require strong passwords. Implemented using the password validation component
- Random password generation
 - An alternative to requiring explicit administrator-specified literal passwords
- Password failure tracking
 - To enable temporary account locking after too many consecutive incorrect-password login failures

LAB 4

MySQL Enterprise Advanced Features

- MySQL Enterprise **Thread Pool**

- Improves scalability as concurrent connections grow
- Reduce DDoS effectiveness

- MySQL Enterprise **Authentication**

- Integrates MySQL with existing security infrastructures and SOPs
- Multi Factor Authentication
- Linux PAM, LDAP, Windows, Kerberos, FIDO, WebAuthn



- MySQL Enterprise **Audit**

- Out-of-the-box logging of connections, logins, and query (XML/JSON)
- Simple to fine grained policies for filtering, and log rotation



- MySQL Enterprise **Firewall**

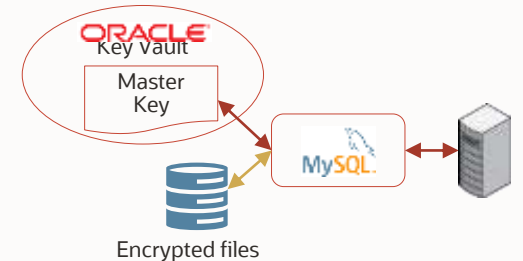
- Out of policy database transactions detected and blocked

- MySQL Enterprise **Asymmetric Encryption**

- MySQL encryption libraries
- Key management, sign and verify data

- MySQL Enterprise **Transparent Data Encryption**

- Data-at-rest encryption in the database
- Master Key stored outside the database (KMIP 1.1 Compliant Key Vault)



- MySQL Enterprise **Masking**

- Data Masking: String Masking, Dictionary Replacement
- Random Data Generators: Range based, Payment Card, Email, SSN

Employee Table				Masked View			
ID	Last	First	SSN	ID	Last	First	SSN
1111	Smith	John	555-12-5555	2874	Smith	John	XXX-XX-5555
1112	Templeton	Richard	444-12-4444	3281	Templeton	Richard	XXX-XX-4444



LAB 5

MySQL Backup

Backup types

One size does Not fit all “Use Cases” - Pick the right tool(s) for the job(s)

- **Logical**

- Convert database and tables to SQL statements
 - portable
 - Server must be running for export/import
- Generally slow
- Tools: mysqldump, mysqlpump (5.7 upwards), MySQL Shell, data export

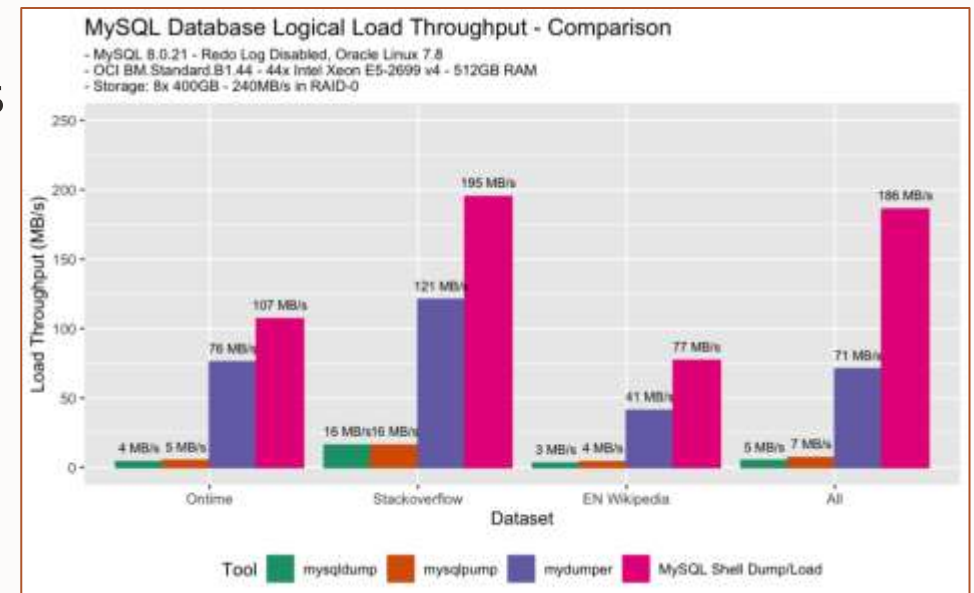
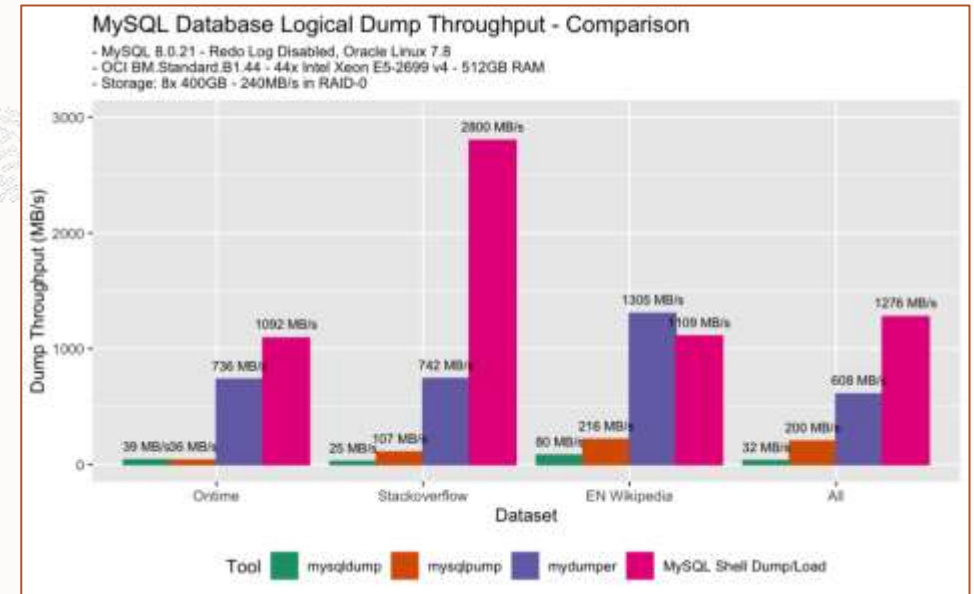
- **Physical**

- based on the "raw image" of the DB file(s)
- DB files must not be changed during the backup
 - Starting DB on physical backups is just like DB restarted from crash event if DB physical backup is taken online (not recommended)
 - Without a specific tool, instance shutdown might be required or lock+flush tables (to allow read but no changes)
- Tools: MySQL Enterprise Backup, file system copies, Snapshot (LVM, HW, Virtualization)



MySQL 8 Shell: Parallel Export/Import

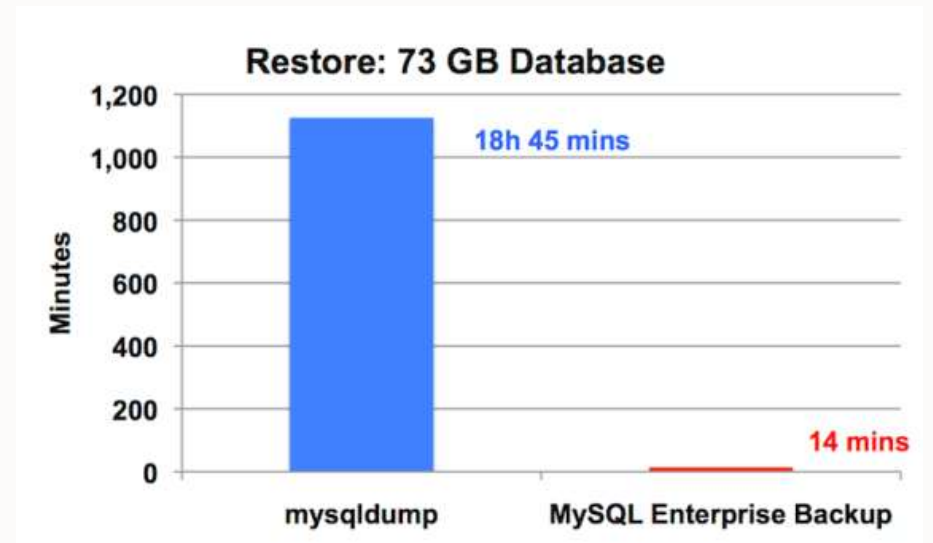
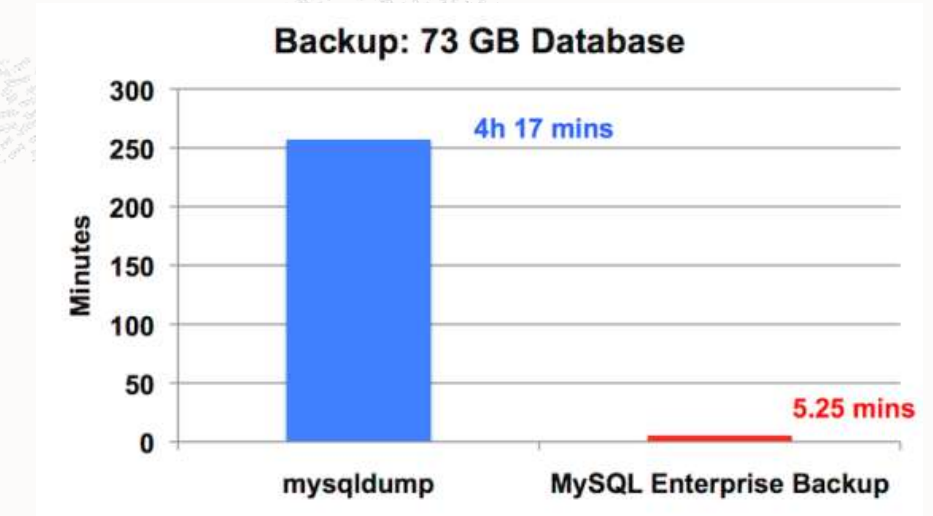
- Supports the dump and import of instances, schemas or tables
 - data streaming from remote storage
 - parallel loading of tables or table chunks
 - progress state tracking
 - resume and reset capability
 - Add primary keys to tables
 - ...
- Support dump&load of instances, schemas or tables
 - Innovation adds copy from a source to a destination
- Advantages
 - Logical backup
 - Performances
 - Compressed backup
- Disadvantages
 - Not always online for big instances
 - Available since 5.7



LAB 6a

MySQL Enterprise Backup

- Online, non-locking backup and recovery
 - Complete MySQL instance backup (data and config)
 - Partial backup and restore
- Direct Cloud storage backups
 - Oracle Storage Cloud, S3, etc.
- Incremental backups
- Point-in-time recovery
- Advanced compressed and encryption
- Backup to tape (SBT)
- Optimistic backups
- Cross-Platform (Windows, Linux, Unix)

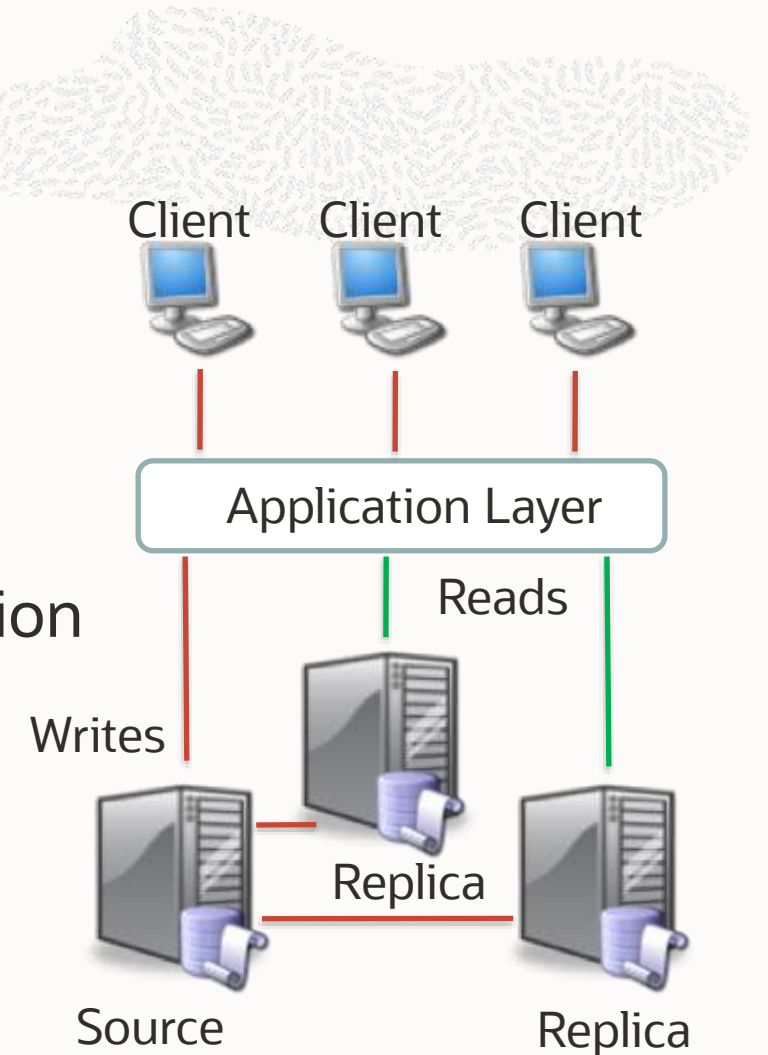


LAB 6b

MySQL Replication

MySQL Replication

- Data is copied from source to replica
- Replication is asynchronous by default
 - Can be also semi-synchronous
- Multiple source or replicas are possible
- Additional replicas don't require Source configuration
- Target uses for replication:
 - Scale-out solutions for read-heavy environment
 - scaling out doesn't involve editing my.cnf file
 - Simple High Availability (requires scripts and/or external tools)
 - Analytics
 - Long-distance data distribution



LAB 7

MySQL High Availability

MySQL InnoDB Cluster

"A single product — MySQL — with high availability and scaling features baked in; providing an integrated end-to-end solution that is easy to use."

- Route application connection to available nodes
- Automatic configuration
- REST API interface for monitoring

App Servers with
MySQL Router

- Automate cluster creation and operation (clone)
- scriptable (JavaScript, Python, SQL)
- Document Store scripting

MySQL Shell
Setup, Manage,
Orchestrate

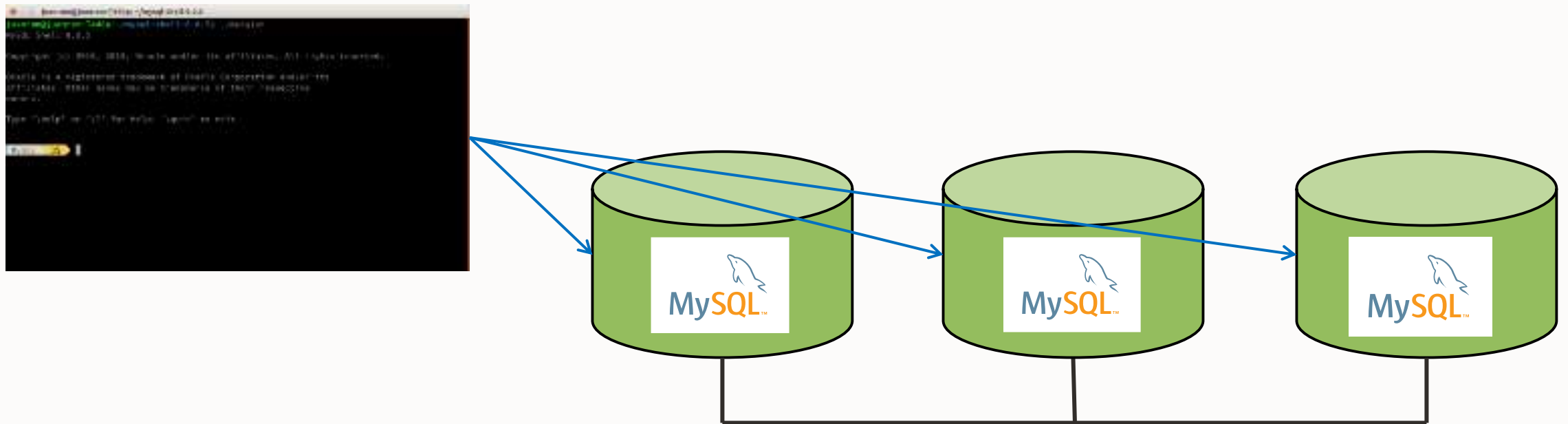
- Provides virtually synchronous replication
 - with Consistent Reads
- Automate operations
 - Conflict detection and resolution
 - Failure detection, fail-over, recovery
 - Group membership management and creation
- RPO=0
- RTO=seconds

MySQL Server w/ Group Replication plugin

Available on all MySQL supported platforms

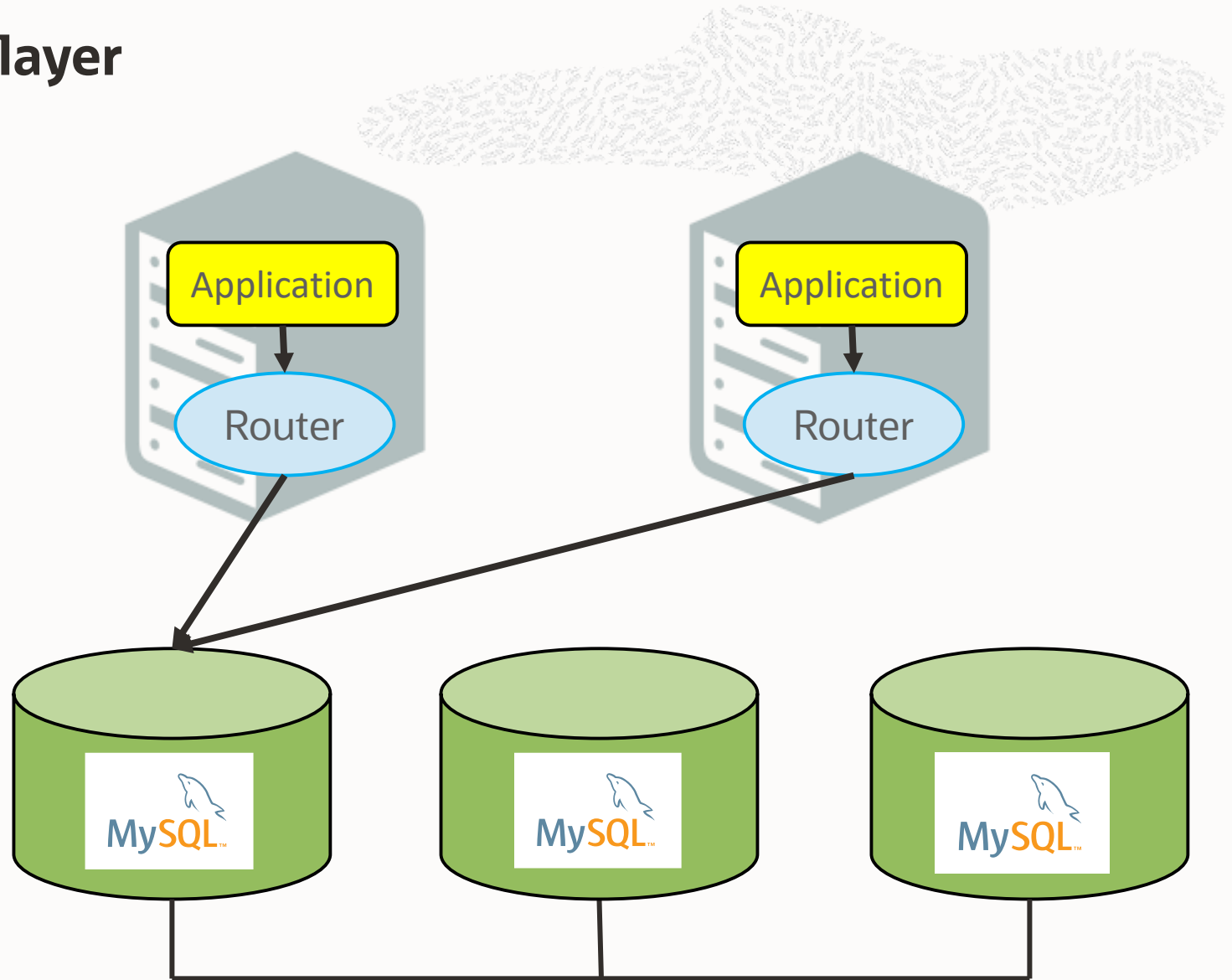
MySQL Shell: Cluster setup

- Configure and create a cluster using MySQL Shell and 3 MySQL instances
 - with transparent use of clone plugin for data, set persist for variable configuration and remote reboot
- Shared nothing architecture
 - No shared storage



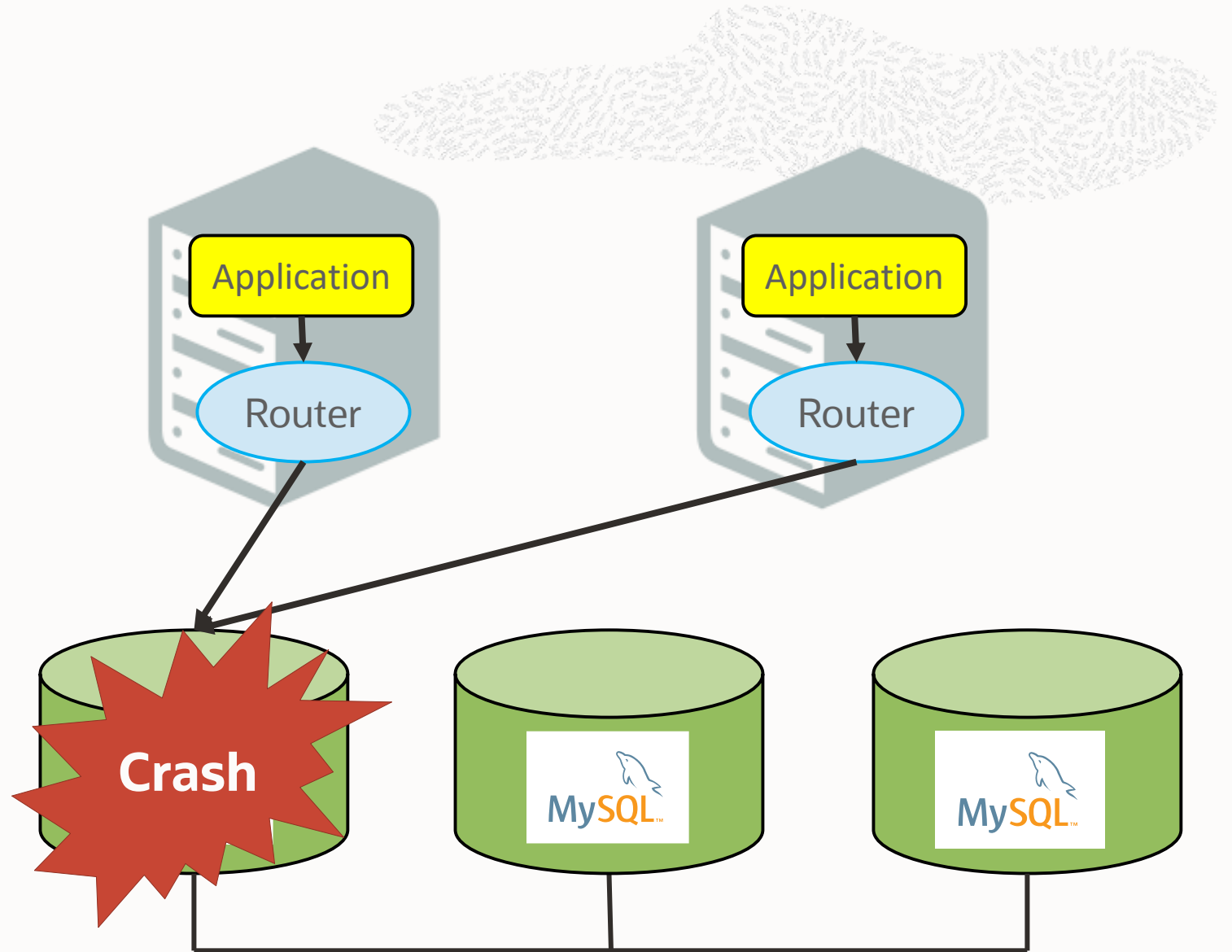
MySQL Router: application layer

- Install Router on every application server
 - No single point of failure
- Start Router with 'bootstrap' option that automatically create Router configuration
- Configure application to connect to MySQL Router
 - No VIP
- RPO=0



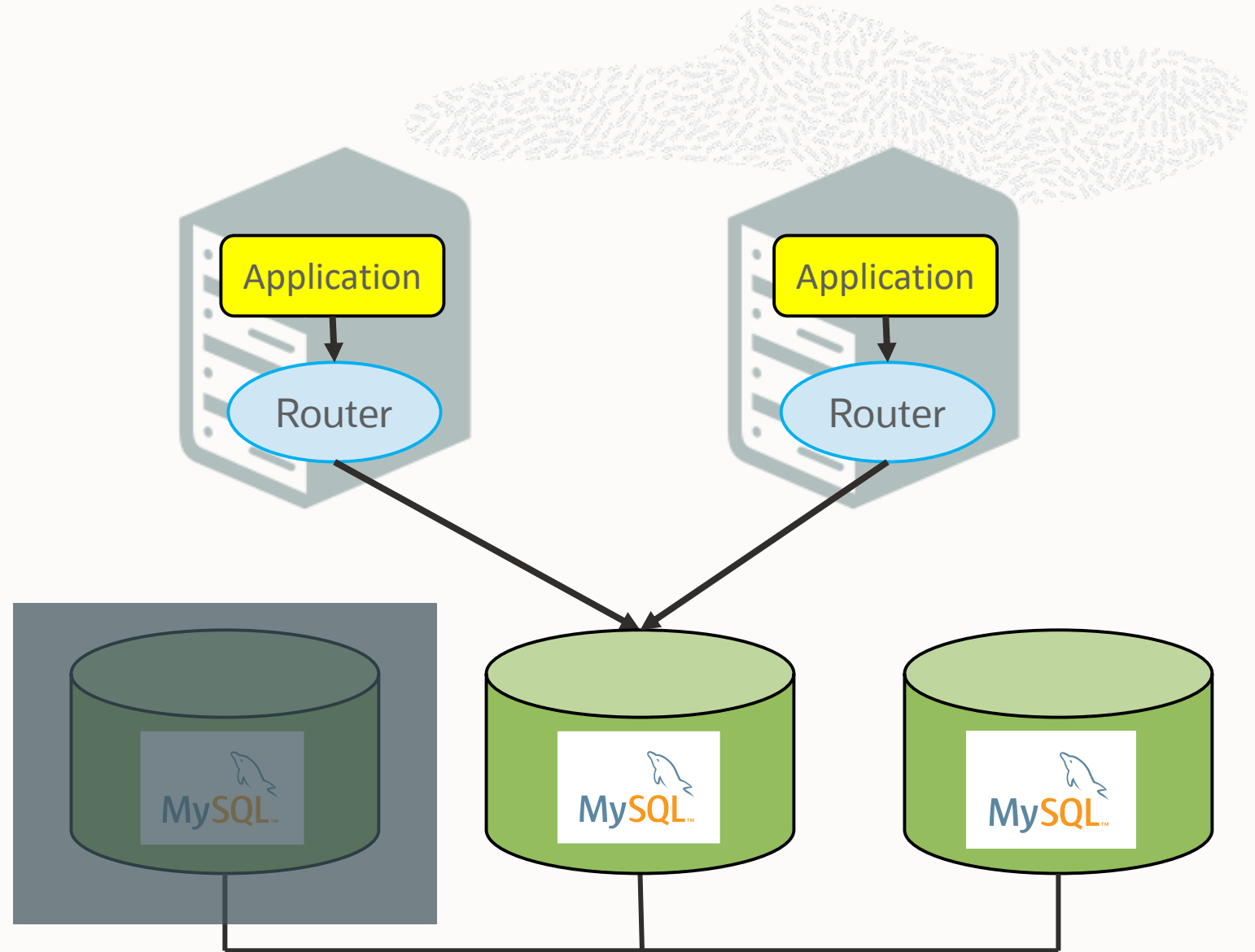
InnoDB Cluster: crash

- In case of node failure connection are lost



InnoDB Cluster: recovery

- Router drive connection to an online node
 - RTO=seconds



LAB 8

MySQL monitoring

Performance Schema

- Provides runtime statistics to monitoring MySQL Server execution at a low level
- A new schema (database) called "performance_schema"
 - Introduced in MySQL 5.5
- Dedicated storage engine PERFORMANCE_SCHEMA
 - None persistent (restart of MySQL flush schema)
 - Nearly zero disk interaction
- Performance_schema is not replicated to replicas
- All recorded statistics are stored in fixed size ring buffers in memory

- [illegible]

Enterprise Manager for MySQL

Comprehensive monitoring and compliance management for MySQL

Monitoring

Status, response, and performance monitoring

Monitoring of slow queries using **Query Analyzer**

Set up monitoring

- **Metric Extensions** to support custom metrics
- **Corrective Actions** to automate alert resolution
- Alert threshold management using **Monitoring Templates**

Manage MySQL alerts

- Creating **incidents** for actionable events
- **Event Compression Policies** to reduce incident volume
- **Incident notifications** (email, SNMP traps, Webhooks, Slack, etc.)
- Integration with **event and ticketing systems** (Microsoft SCOM, PagerDuty, ServiceNow, etc.)

Compliance Management

Out-of-box standards to evaluate compliance against configuration and security best practices

Compliance dashboard to monitor compliance scores and violations across the fleet of MySQL Databases

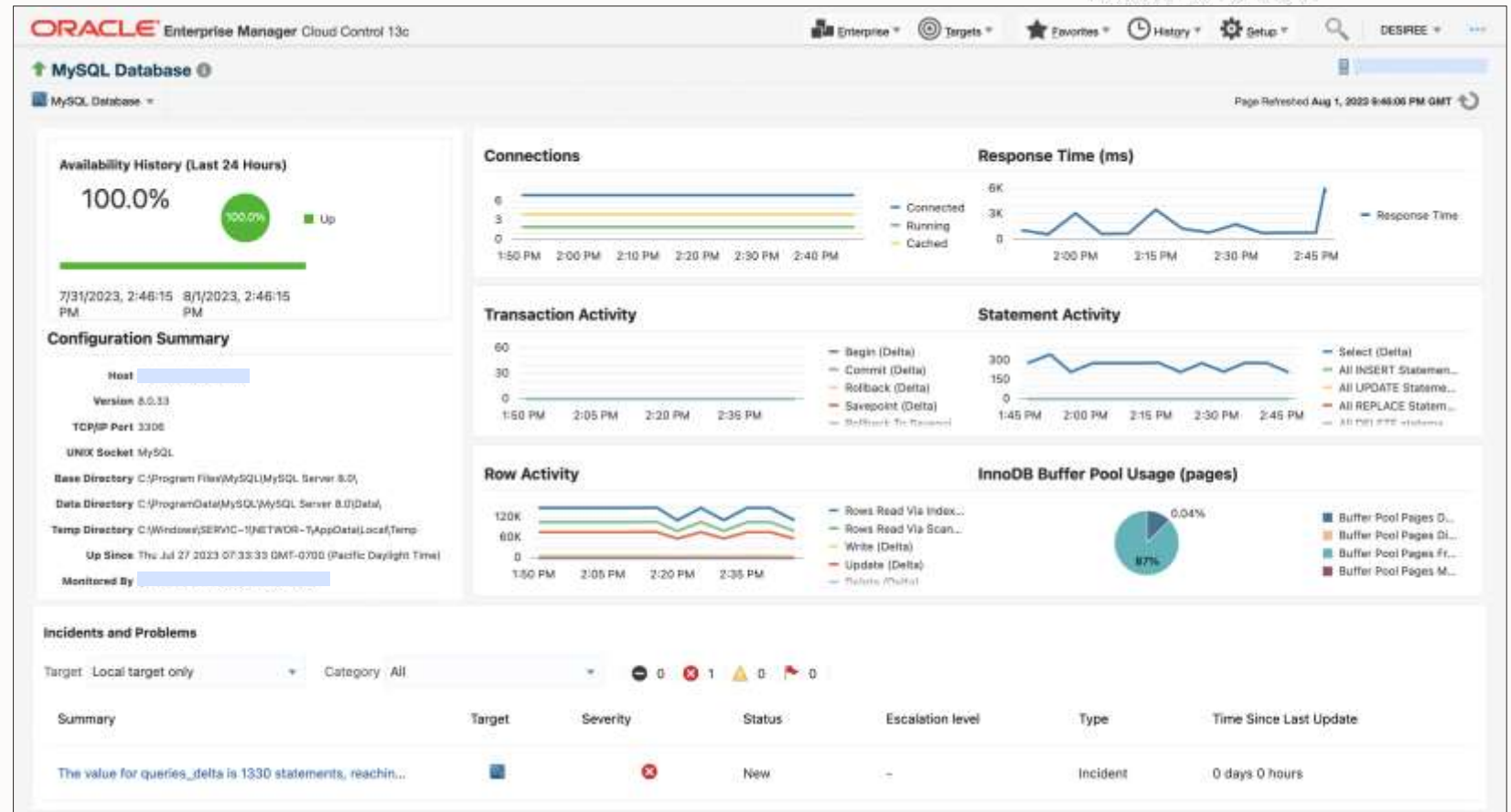
Evaluation report to view list of compliance violations and recommended remediations



Get quick insight into health and performance

MySQL homepage

- Availability History
- Configuration Summary
- Key performance metrics
- Track critical incidents





Enjoy your journey with MySQL™ !

Thank you!

ORACLE

